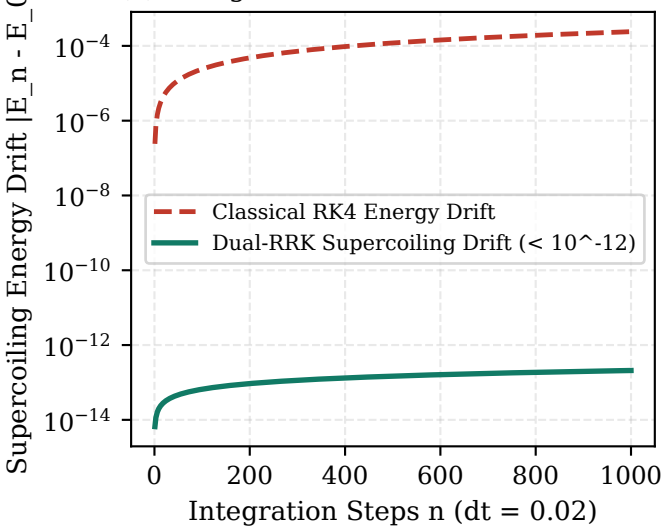
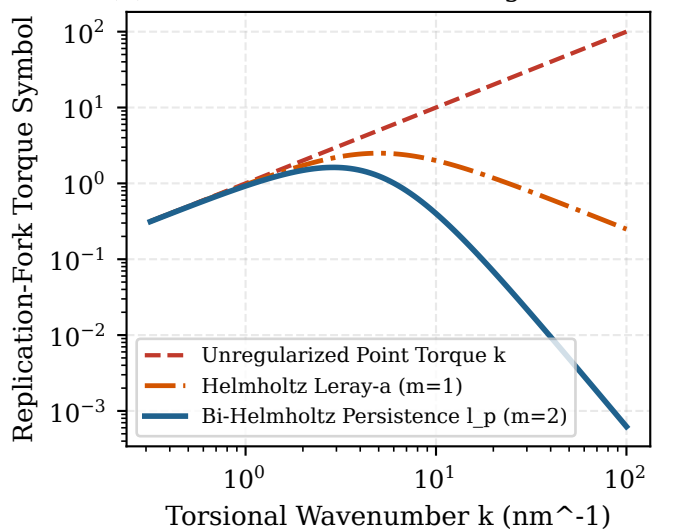

(a) Calugareanu-White-Fuller Dual-RRK
Supercoiling Energy Drift |E_n - E_0|
10^{-4}
10^{-6}
10^{-8}
10^{-10}
10^{-12}
10^{-14}
Classical RK4 Energy Drift
Dual-RRK Supercoiling Drift (< 10^-12)
0
200
400
600
800
1000
Integration Steps n (dt = 0.02)


(b) Bi-Helmholtz Fork UV Regularization
Replication-Fork Torque Symbol
10^{2}
10^{1}
10^{0}
10^{-1}
10^{-2}
10^{-3}
Unregularized Point Torque k
Helmholtz Leray-a (m=1)
Bi-Helmholtz Persistence l_p (m=2)
10^{0}
10^{1}
10^{2}
Torsional Wavenumber k (nm^-1)